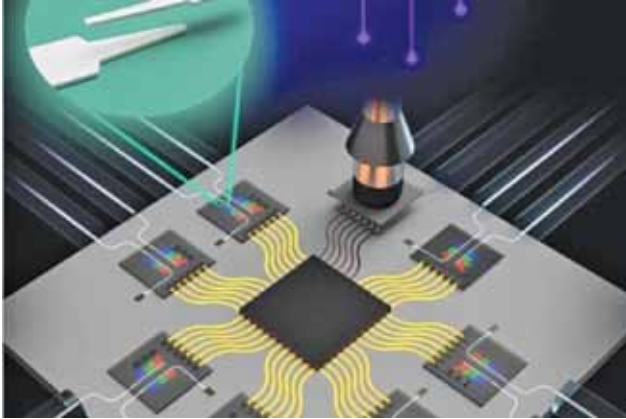




Breakthrough in co-packaging photonic and electronic chips

The MIT (United States) FUTUR-IC team has developed a breakthrough chip packaging method that co-integrates electronics and photonics using passive alignment. This feature eliminates the need for costly laser tuning and allows automated, high-volume assembly using existing foundry tools.



The MIT device in the green callout could be key to faster, more energy-efficient data communication. It solves a major problem associated with packaging an electrical chip (black, center) with photonic chips (the eight surrounding squares). This image also shows an automated tool placing the final photonic chip into position (Credit: Drew Weninger, MIT)

The benefit is faster, cheaper, and more energy-efficient data systems. With evanescent couplers enabling vertical light transfer and relaxed alignment tolerance, this innovation supports scalable, multilayer designs vital for the next-gen computing and communication infrastructure.

Thin-film sensors that operate from -50° to 950°C without protective coatings

A team at Zhejiang University (Zhejiang, China) has developed a one-step laser method to create thin-film temperature sensors that operate from -50°C to 950°C , without the need for protective coatings. A single laser pulse forms both the sensing and oxidation-resistant layers, simplifying production, reducing material usage, and enabling direct application on curved, high-heat surfaces such as engines or pipelines. These sensors deliver real-time data with minimal signal drift, enhancing predictive maintenance.

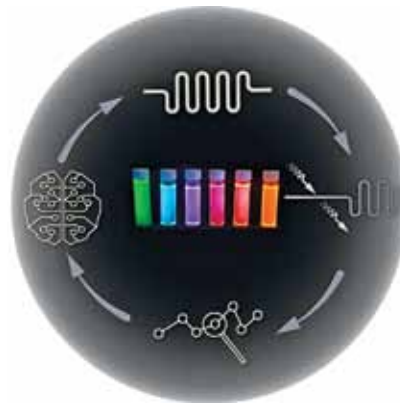
Compact metal-free thin-film supercapacitor delivers 200V

GDUT (Guangzhou, China) researchers have developed a metal-free thin-film supercapacitor (TFSC) stack that delivers 200V in just 3.8cm^3 . Using a CO_2 laser, they converted polyimide paper into multifunctional 3D graphene, serving

as an electrode, conductor, and support. Stacking 160 mini TFSCs with a hydrogel electrolyte, the design eliminates metal connectors, slashes bulk, and suits extreme environments, making it ideal for wearables, flexible sensors, and next-generation electronics, says Prof. Huilong Liu.

AI-powered self-driving lab tests materials 10x faster

Researchers at NC State (Raleigh, North Carolina) have developed an AI-powered self-driving lab that uses dynamic-state flow and real-time data to test materials 10x faster than



A cutting-edge self-driving lab now uses real-time experiments to gather data continuously, making materials discovery 10x faster and far more efficient (Credit: Milad Abolhasani, North Carolina State University)

traditional labs. Unlike slow, sequential testing, this lab runs continuous experiments guided by a smart ML engine. It identified the best CdSe quantum dot on its first try. This breakthrough cuts costs, slashes waste, and accelerates discovery for clean energy, electronics, and sustainable materials, within just days.

Quantum batteries get a big storage time boost

Researchers at RMIT (City of Melbourne, Australia) have built a prototype that stores energy 1000 times longer than



Quantum battery that can store energy for microseconds (Credit: RMIT University)

previous versions, thanks to a phenomenon called superabsorption. The team used triplet states of electrons and layered materials, Rhodamine 6G dye, a palladium compound, and a polymer separator, to trap energy from a green laser. Though it lasts just microseconds, this

breakthrough hints at ultra-fast charging and long-lasting energy storage.